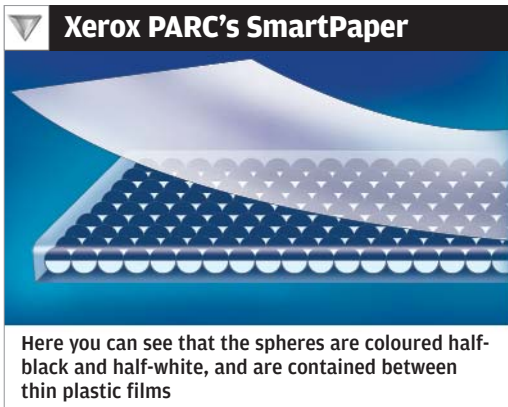




Infographic: Shaktishima Patkar

research on ePaper. To give you an idea of how ahead of his time Sheridan was, the novel idea back then was to replace paper with portable flat-panel displays, which, as we all know, were not exactly portable back then, were very expensive, and had terrible contrast—you could hardly make out the screen in a lit room, forget about outdoors!

Sheridan's dream was to make a flexible paper-like sheet that could not only display text and images, but also be re-written by using an electrostatic charge. The solution he came up with was to use tiny electrically-charged spheres, enclosed in the oil-filled cavities of a thin layer of a transparent elastomer.

Incidentally, this is the technology Xerox PARC still uses in its research of ePaper, which they call SmartPaper. Each 100-micron sphere is coloured half white and half black (or another dark colour). The spheres contain a dipole—like a magnet, with positive and negative charges on opposite ends.

Thus, when an electric charge is passed over certain pre-designated areas of the elastomer film, the spheres in those areas rotate to show their dark side, colouring the area. A chip can control the charges at different areas to produce a sort of advanced, electronically-controlled etch-a-sketch!

Since the spheres are reflective in nature, they are extremely high-contrast, making them readable even under direct sunlight. This technology has already been used to make some interesting products. We will talk about those a little later. Needless to say, this technology, though easily implementable, has the drawback of being monochrome. Still, it's a step in the right direction.

The Leaders

Say the words "paper display" and the name that springs to mind is E-Ink Corporation—the Xerox or Discman of paper displays! A pioneer in this field, E-Ink is consistently working towards building better and slimmer displays.

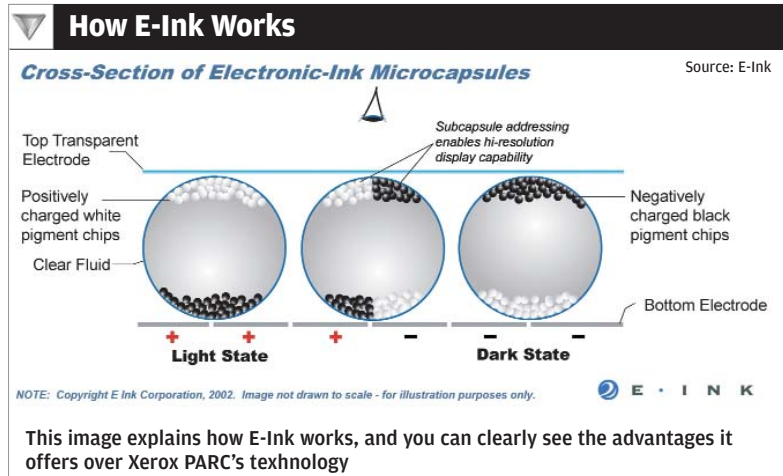
E-Ink currently provides technology and paper displays to many corporates—not just to help them build prototypes, but also for full-fledged products. E-Ink's investors include Toppan Printing of Japan, Intel Capital Corporation, Motorola Corporation, Philips Components (a division of Royal Philips Electronics) and Vivendi Universal Publishing (France).

The list of companies that are actively researching paper displays is even longer. Needless to say, the market is waiting impatiently to see some significant progress in this field, and researchers are working overtime to meet this demand.

E-Ink's technology is similar to what Sheridan came up with at Xerox PARC, but is definitely not the same. Instead of using microcapsules that are coloured half-black and half-white, E-Ink uses microcapsules that contain granules of a positively-charged white pigment and granules of a negatively-charged black pigment.

These capsules are suspended in a clear fluid that lies between thin layers of a plastic material, which is, in turn, laminated on a layer of micro-circuitry and electrodes. The top plastic layer is a transparent electrode, whereas the bottom layer contains the control electrodes.

Now, in order to draw a display, the electrodes are charged either positive or negative, according to the image requirement. The



micro-circuitry is small enough for two electrodes to be placed under each microcapsule, and if both electrodes are charged positively, the positively-charged white pigment granules in the capsule directly above them are pushed to the top, thus forming a white dot, or pixel. Similarly, a negative charge causes the negatively-charged black pigment granules to be pushed up, thus forming a black pixel.

Now, the two electrodes under each microcapsule can be polarised differently as well, thus forming a half-white and half-black pixel. This is how E-Ink's technology achieves higher resolutions and apparent shades of grey. All this circuitry forms a screen with a calculable resolution, which can then be controlled by a display driver or graphics chip.

Moreover, the "E-Ink" is reflective in nature, and it looks just like print on paper, has a readable angle of 170 degrees, and is clear even in direct sunlight.

But enough on costs and technology briefs—let's get on to the actual prototypes that have been developed to further illustrate the road ahead.